🌍 Project: CarbonPulse AI

A Cyber-Themed Carbon Footprint Intelligence Platform

Built for Hack2Skills PromptWars Challenge 3

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1. MongoDB Data Structure (Optimized Mongoose Schemas)

User Schema

Stores profile, calculated baseline footprint, streaks, achievements, and gamification metrics.

```
// models/User.js

import mongoose from "mongoose";

const UserSchema = new mongoose.Schema(
{
  name: {
    type: String,
    required: true,
    trim: true,
    maxlength: 50
  },

  email: {
    type: String,
    required: true,
    unique: true,
    lowercase: true
  },

  avatar: {
    type: String,
    default: ""
  },

  baselineFootprint: {
    type: Number,
```

```
    default: 0
  },

  currentFootprint: {
    type: Number,
    default: 0
  },

  points: {
    type: Number,
    default: 0
  },

  level: {
    type: Number,
    default: 1
  },

  streak: {
    type: Number,
    default: 0
  }
},
{
  timestamps: true
}
);

/*
INDEX STRATEGY

email → login lookup
createdAt → leaderboard sorting

*/

UserSchema.index({ email: 1 });
UserSchema.index({ createdAt: -1 });

export default mongoose.models.User ||
mongoose.model("User", UserSchema);
```

Activity Log Schema

Tracks daily user actions.

```
// models/ActivityLog.js

import mongoose from "mongoose";

const ActivityLogSchema = new mongoose.Schema(
{
  userId: {
    type: mongoose.Schema.Types.ObjectId,
    ref: "User",
    required: true
  },

  date: {
    type: Date,
    default: Date.now
  },

  transport: {
    mode: {
      type: String,
      enum: [
        "walk",
        "cycle",
        "bus",
        "train",
        "car",
        "bike"
      ]
    },

    distanceKm: {
      type: Number,
      default: 0
    }
  },

  diet: {
    type: String,
    enum: [
      "vegan",
      "vegetarian",
      "mixed",
      "high_meat"
    ]
  },

  microActions: [
    {
```

```
      action: String,
      carbonSavedKg: Number
    }
  ],

  totalCarbonSaved: {
    type: Number,
    default: 0
  }
},
{
  timestamps: true
}
);

/*
INDEX STRATEGY

userId + date

Fast dashboard retrieval
Fast analytics queries
Fast progress history

*/

ActivityLogSchema.index({
  userId: 1,
  date: -1
});

export default mongoose.models.ActivityLog ||
mongoose.model("ActivityLog", ActivityLogSchema);
```

Carbon Target Schema

Stores sustainability goals.

```
// models/CarbonTarget.js

import mongoose from "mongoose";

const CarbonTargetSchema = new mongoose.Schema(
{
  userId: {
```

```
    type: mongoose.Schema.Types.ObjectId,
    ref: "User",
    required: true
  },

  type: {
    type: String,
    enum: ["weekly", "monthly"],
    required: true
  },

  targetReductionKg: {
    type: Number,
    required: true
  },

  achievedReductionKg: {
    type: Number,
    default: 0
  },

  progressPercentage: {
    type: Number,
    default: 0
  },

  startDate: Date,
  endDate: Date,

  status: {
    type: String,
    enum: [
      "active",
      "completed",
      "expired"
    ],
    default: "active"
  }
},
{
  timestamps: true
}
);

/*
INDEX STRATEGY

userId + status
```

```
Instant loading of active goals

*/

CarbonTargetSchema.index({
  userId: 1,
  status: 1
});

export default mongoose.models.CarbonTarget ||
mongoose.model("CarbonTarget", CarbonTargetSchema);
```

Lightweight Architecture (<10MB Repo)

```
Frontend
|
├── Next.js
├── Tailwind CSS
├── Recharts (CDN Import)
|
Backend
|
├── Next.js API Routes
├── MongoDB Atlas
├── Mongoose
|
AI Layer
|
├── Gemini API
├── OpenAI API
├── Claude API
|
External Carbon Data
|
├── Climatiq API
├── Carbon Interface API
|
Hosting
|
├── Vercel
└── MongoDB Atlas
```

No local datasets.
No image assets.
No component libraries.
No heavy UI frameworks.

---

2. Cyberpunk UI/UX Configuration

---

tailwind.config.ts

```
import type { Config } from "tailwindcss";

const config: Config = {
  content: [
    "./app/**/*.{js,ts,jsx,tsx,mdx}",
    "./components/**/*.{js,ts,jsx,tsx}"
  ],

  theme: {
    extend: {
      colors: {
        cyber: {
          bg: "#060816",
          card: "#0B1023",

          cyan: "#00F5FF",
          purple: "#A855F7",

          success: "#22C55E",
          warning: "#FACC15",
          danger: "#EF4444",

          text: "#E5E7EB",
          muted: "#94A3B8"
        }
      },

      boxShadow: {
        cyanGlow:
          "0 0 10px rgba(0,245,255,.5), 0 0 20px rgba(0,245,255,.3)",

        purpleGlow:
          "0 0 10px rgba(168,85,247,.5), 0 0 20px rgba(168,85,247,.3)"
      },
```

```
      backgroundImage: {
        cyberGrid:
          "linear-gradient(rgba(255,255,255,0.03) 1px, transparent 1px),linear-gradient(90deg, rgba(255,255,255,0.03) 1px, transparent 1px)"
      }
    }
  },

  plugins: []
};

export default config;
```

---

global.css

```
@tailwind base;
@tailwind components;
@tailwind utilities;

html,
body {
  background: #060816;
  color: #e5e7eb;
  min-height: 100vh;
}

body {
  background-image:
    linear-gradient(
      to bottom,
      rgba(0,245,255,.03),
      transparent
    );
}

* {
  border-color: rgba(255,255,255,.08);
}

.glow-cyan {
  box-shadow:
    0 0 8px rgba(0,245,255,.6),
    0 0 20px rgba(0,245,255,.3);
}
```

```
.glow-purple {
  box-shadow:
    0 0 8px rgba(168,85,247,.6),
    0 0 20px rgba(168,85,247,.3);
}

.cyber-card {
  background: rgba(11,16,35,.85);
  backdrop-filter: blur(12px);
  border: 1px solid rgba(255,255,255,.08);
  border-radius: 16px;
}

.neon-text {
  text-shadow:
    0 0 10px rgba(0,245,255,.8);
}
```

Recommended Dashboard Layout

```
----------------------------------------------
CARBONPULSE AI
----------------------------------------------

[ Footprint Score ]

[ Carbon Saved ] [ Weekly Goal ]

---------------------------------------------
AI Insights
---------------------------------------------

"Switching to cycling twice a week
can reduce 13kg CO₂ monthly."

---------------------------------------------
Daily Actions
---------------------------------------------

☑ Reusable Bottle
☑ Public Transport
☑ Vegetarian Meal

---------------------------------------------
```

Leaderboard

#1 User A
#2 User B
#3 User C

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3. README.md

CarbonPulse AI 🌍⚡

Overview

CarbonPulse AI is a lightweight cyber-themed sustainability platform that helps users understand, track, and reduce their carbon footprint through personalized recommendations, daily action tracking, AI-generated insights, and gamified progress systems.

Built for Hack2Skills PromptWars Challenge 3.

Problem Statement

Millions of people want to live sustainably but lack visibility into how their daily actions affect the environment.

Existing carbon calculators are often static, complex, and fail to motivate long-term behavioral change.

CarbonPulse AI transforms carbon awareness into an engaging, actionable, and rewarding experience.

Key Features

Carbon Footprint Tracking

Track:

- Transportation habits

- Dietary choices
- Daily eco-friendly actions

AI Sustainability Coach

Provides personalized recommendations using AI.

Examples:

- Switch two short car trips per week to cycling.
- Reduce meat consumption one day weekly.
- Optimize household energy usage.

Goal Management

- Weekly targets
- Monthly targets
- Progress analytics

Gamification

- Eco Points
- Achievement Levels
- Daily Streaks
- Community Leaderboards

AI Multi-Agent Workflow

The project was designed using multiple AI systems:

Gemini

- Sustainability research
- Carbon reduction datasets
- Environmental insight generation

ChatGPT

- Software architecture
- Database design
- UI/UX strategy
- Tailwind system design

Claude

- Logic optimization

- Data modeling
- API integration planning

Jules

- Autonomous coding assistance
- Refactoring
- Boilerplate generation

Technology Stack

Frontend

- Next.js
- Tailwind CSS
- Recharts

Backend

- Next.js API Routes
- MongoDB Atlas
- Mongoose

AI Services

- Gemini API
- OpenAI API
- Claude API

Carbon APIs

- Climatiq API
- Carbon Interface API

Deployment

- Vercel
- MongoDB Atlas

Lightweight Repository Strategy (<10MB)

To comply with the repository size limitation:

- No local datasets
- No image assets

- No component libraries
- No UI kits
- No heavy animation libraries
- No local AI models

All intelligence is powered through cloud APIs and lightweight serverless architecture.

Repository size remains extremely small while functionality remains enterprise-grade.

Installation

git clone

cd carbonpulse-ai

npm install

Create .env.local

MONGODB_URI=

GEMINI_API_KEY=

OPENAI_API_KEY=

CLAUDE_API_KEY=

npm run dev

Open:

http://localhost:3000

Future Scope

- Carbon footprint prediction
- Smart IoT integrations
- AI sustainability assistant
- ESG reporting dashboard
- Community challenges

Team

Joe Daniel

B.Tech Artificial Intelligence & Data Science

KGISL Institute of Technology

🎤 2-Minute Hackathon Pitch

Good morning judges,

My name is Joe Daniel, and I am a second-year B.Tech Artificial Intelligence and Data Science student from KGISL Institute of Technology.

Today, I am presenting CarbonPulse AI, a cyber-themed intelligent platform designed to help individuals understand, track, and reduce their carbon footprint through simple daily actions.

One of the biggest challenges in sustainability is that people want to help the environment, but they don't know how their everyday choices impact carbon emissions. Most carbon calculators simply provide a number and stop there.

CarbonPulse AI goes beyond calculation.

Our platform tracks transportation habits, dietary choices, and eco-friendly activities, then uses AI to generate personalized recommendations that users can immediately act upon. We transform sustainability into a rewarding experience through goals, streaks, points, and leaderboards.

What makes CarbonPulse AI unique is its futuristic cyberpunk-inspired interface. Neon cyan and purple visualizations make environmental data engaging and intuitive rather than overwhelming.

From a technical perspective, one of our biggest achievements was designing the entire system under a strict 10MB repository limit. Instead of relying on large datasets, local AI models, or bulky UI frameworks, we built a highly optimized architecture using Next.js, MongoDB Atlas, cloud APIs, and serverless services. This approach delivers powerful functionality while keeping the repository extremely lightweight and deployment-ready.

Additionally, the project was developed using a multi-agent AI workflow involving Gemini, ChatGPT, Claude, and Jules, allowing rapid research, architecture design, logic optimization, and autonomous code generation.

CarbonPulse AI demonstrates that impactful climate technology does not need heavy infrastructure. With intelligent design and efficient architecture, we can make sustainability accessible, engaging, and scalable.

Thank you.

🏆 Hackathon Winning Angle

Your differentiator should be:

"AI-Powered Carbon Intelligence + Cyberpunk Gamification + Under 10MB Architecture."

Most teams will build a calculator. You are building an **interactive sustainability operating system** that feels modern, intelligent, and technically impressive while meeting the repository constraint elegantly.